

For the voltmeter to record zero voltage, the ratio of the resistance $P/Q = R/S$

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Ans: C

2018 JC2 H2 Physics Prelims Exam Solution (**9749** Paper 1)

*Current along PR is parallel to the magnetic field and so there is no magnetic force on PR.
The current along QP has a component flowing vertically downwards and perpendicular to the magnetic field. Using Fleming's LHR, the force acts on wire QP out of the page.
The current along RQ has a component flowing vertically upwards and perpendicular to the magnetic field. Using Fleming's LHR, the force acts on wire RQ into the page.
Considering the two forces on the triangular wire QPR, the triangle rotates about the axis **WX** with R pointing into the page.*